

Brain Positron Emission Tomography (PET) and PET-CT

ACG: A-0096 (AC)

[Link to Codes](#)

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Clinical Indications

- Brain positron emission tomography (PET) and PET-CT may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - Autoimmune encephalitis, suspected, as indicated by **ALL** of the following(29)(30):[N](#)
 - Limbic involvement suspected, as indicated by **1 or more** of the following(31):
 - Behavioral changes (eg, aggression, agitation, disinhibition)
 - Memory deficits
 - Psychiatric symptoms (eg, anxiety, depression, mood changes, psychosis)
 - Seizures
 - Sleep changes (eg, insomnia, hypersomnia)
 - MRI of brain inconclusive, negative, or contraindicated
 - Dementia or mild cognitive impairment, as indicated by **1 or more** of the following(32)(33)(34)(35)(36):[N](#)
 - Need to distinguish between Alzheimer disease and other types of dementia (eg, frontotemporal dementia, dementia with Lewy bodies), and **ALL** of the following:
 - Other clinical (eg, neurocognitive tests) and imaging tests have been inconclusive.
 - Results will impact treatment plan.
 - Patient with mild cognitive impairment or mild dementia on neurocognitive testing, and results will impact treatment plan
 - Epilepsy, with need for presurgical imaging, as indicated by **ALL** of the following(62)(63)(64)(65)(66):[N](#)
 - Focal epilepsy^(A)(86)
 - Poor or no response to 2 or more anticonvulsant medications at maximal dose (ie, refractory or drug-resistant epilepsy) (87)(88)(89)
 - Surgery planned or being considered(80)(90)
 - Results of brain MRI negative, indeterminate, or discordant with other testing for seizure focus(89)
 - Primary brain cancer evaluation needed, as indicated by **1 or more** of the following(91)(92)(93)(94):[N](#)
 - Post-treatment evaluation needed to differentiate residual or recurrent tumor from treatment effects (eg, radiation injury or "pseudoprogression" from radiation and temozolomide)(9)(105)(106)(107)(108)
 - Preoperative planning needed for tumor resection by defining tumor heterogeneity and extension
 - Primary brain tumor, suspected, with indeterminate MRI(109)
 - Symptoms suggestive of Parkinson disease or other neurodegenerative disorder(110)(111):[N](#)

- Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

PET is a scintigraphic imaging technique providing detailed information about localized metabolism of labeled trace compounds, most commonly F-18 fluorodeoxyglucose as an analogue for glucose metabolism.(3)(6)(7)(8) **(EG 2)** A number of PET radiopharmaceuticals other than F-18 fluorodeoxyglucose that are intended to enhance diagnostic sensitivity and specificity in imaging primary brain tumors such as gliomas have been introduced or are in development.(9)(10)(11)(12) **(EG 2)** Amyloid imaging agents, including ¹⁸F-florbetaben, ¹⁸F-florbetapir, and ¹⁸F-flutemetamol, have been introduced to improve identification of areas of the brain with large amounts of amyloid-beta, a pathologic protein.(13)(14)(15)(16)(17)(18)(19) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 102 published peer reviewed articles, 7 specialty society or other evidence-based guidelines, and 4 Cochrane systematic reviews.

For autoimmune encephalitis, suspected, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A consensus-based guideline for the diagnosis of autoimmune encephalitis notes that, for the evaluation of limbic autoimmune encephalitis, PET imaging may be used either as a substitute for brain MRI as or an ancillary diagnostic test if brain MRI results are inconclusive.(29) **(EG 2)** A narrative review notes that metabolic patterns found on brain F-18 fluorodeoxyglucose PET may assist in the diagnosis of autoimmune encephalitis, particularly when MRI results are negative or inconclusive.(30) **(EG 2)**

For dementia or mild cognitive impairment, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines recommend brain MRI as first-line imaging for diagnosis.(37)(38)(39) **(EG 2)** F-18 fluorodeoxyglucose PET scan can confirm Alzheimer disease and differentiate it from frontotemporal lobar degeneration or other dementias,(40)(41)(42) and with an appropriate tracer, PET scan can help diagnose mild cognitive impairment due to Alzheimer disease.(32)(43)(44)(45)(46) **(EG 2)** However, for patients with mild cognitive impairment, a meta-analysis demonstrated that F-18 fluorodeoxyglucose PET lacked the ability to predict which people with mild cognitive impairment would develop Alzheimer disease or other dementia over a period of time.(47) **(EG 1)** Expert opinion is divided as to whether F-18 fluorodeoxyglucose PET scan should be included as a routine diagnostic procedure for dementia.(48)(49)(50)(51) **(EG 2)** Cohort studies of patients with preclinical and early Alzheimer disease have shown that F-18 fluorodeoxyglucose PET scan is able to detect evidence of disease; the information was redundant when compared with MRI morphometry results.(52)(53) **(EG 2)** An expert consensus guideline proposes amyloid PET for assessment of amyloid-beta proteinopathy, one of the core biomarkers of Alzheimer disease, and suggests that amyloid PET be used for diagnosis of Alzheimer disease and for treatment selection.(54) **(EG 2)** Systematic reviews investigating PET scan use of amyloid imaging markers (eg, ¹⁸F-florbetaben, ¹⁸F-florbetapir, ¹⁸F-flutemetamol) for the detection of patients with mild cognitive impairment who will convert to Alzheimer disease concluded that these markers were not appropriate for routine clinical use due to varying or poor sensitivity and specificity, as well as the small numbers of patients in the trials.(14)(15)(16) **(EG 1)** A cross-sectional study of 719 patients recruited from 3 memory disorder clinics found that brain PET scan with ¹⁸F-flortaucipir, an investigational tau imaging agent, had sensitivity of 89.9% and specificity of 90.6% for differentiating Alzheimer disease from other neurodegenerative diseases; furthermore, ¹⁸F-flortaucipir PET was superior to volumetric MRI measures for discriminating between Alzheimer dementia and other neurodegenerative dementias.(55) **(EG 2)** A prospective study of tau PET imaging (using ¹⁸F-flortaucipir or ¹⁸F-RO948) as a prognostic marker for Alzheimer dementia evaluated 1431 patients (315 patients with Alzheimer dementia with amyloid on PET and/or cerebrospinal fluid, 443 patients with mild cognitive impairment (amyloid positive or negative), and 673 cognitively unimpaired patients (amyloid positive or negative)) and found, at a mean follow-up of 1.9 years, that tau PET imaging more accurately predicted decline in cognitive function (measured with Mini-Mental State Examination) compared with amyloid PET imaging or MRI across all patient groups.(56) **(EG 2)** An observational study of 94 patients with clinical Alzheimer dementia evaluated with ¹⁸F-flutemetamol PET scan found no association between measured amyloid burden and clinical deterioration (measured by change in the Clinical Dementia Rating Scale Sum of Boxes) over time.(57) **(EG 2)** A comparative effectiveness systematic review notes that for distinguishing Alzheimer from non-Alzheimer dementia, the sensitivity and specificity are 91% and 92%, respectively, for amyloid PET scan and 89% and 74%, respectively, for fluorodeoxyglucose PET scan.(58) **(EG 2)** Although amyloid imaging markers have been validated as biomarkers, more research is needed to establish their role in diagnosis, patient management, and improving clinical outcomes.(14)(15)(16)(59)(60)(61) **(EG 2)**

For epilepsy, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review of studies using neuroimaging for drug-resistant temporal lobe epilepsy found that in patients with an unremarkable brain MRI, an

abnormal brain PET scan was a favorable prognostic indicator for anterior temporal lobectomy.(67) **(EG 1)** An expert consensus guideline recommends MRI as the first-line imaging option for preoperative evaluation of refractory epilepsy(63); however, 15% to 30% of patients with refractory focal epilepsy have negative findings of seizure focus on brain MRI.(68) **(EG 2)** For preoperative localization, brain PET can demonstrate a seizure focus as an area of hypometabolism on interictal (between seizure episodes) imaging, and is particularly helpful when MRI is normal or is discordant with findings on ictal (during seizure) EEG.(69)(70)(71)(72)(73) **(EG 2)** Ictal SPECT can also localize the epileptic focus when MRI has been unrevealing.(69)(74)(75) **(EG 2)** Review articles and cohort studies note that both PET and SPECT can be useful for the presurgical evaluation of patients with temporal and extratemporal lobe epilepsy.(65)(72)(76)(77) **(EG 2)** Review articles state that subtraction ictal SPECT can be useful for presurgical evaluation of pediatric patients with West syndrome (an age-dependent epileptic encephalopathy).(78)(79) **(EG 2)** Review articles note that a major role of brain SPECT is to help localize the epileptogenic zone during the presurgical evaluation of focal epilepsy that has been shown to be pharmacoresistant.(73)(80) **(EG 2)** The coregistration of subtraction ictal SPECT with MRI substantially increases the predictive value of good surgical outcome to greater than 55% seizure free after focal epilepsy surgery.(72)(81)(82)(83) **(EG 2)**

For primary brain cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A meta-analysis of 26 studies concluded that both F-18 fluorodeoxyglucose and C-11 methionine PET scan have adequate accuracy for detecting recurrent glioma when suspected on CT scan or MRI.(95) **(EG 2)** A meta-analysis of 30 studies concluded that F-18 fluorodeoxyglucose PET scan had pooled sensitivity of 71% and pooled specificity of 80% for differentiation of brain tumors; C-11 methionine PET scan had pooled sensitivity of 91% and pooled specificity of 86%.(96) **(EG 1)** A meta-analysis of 23 studies (994 patients) concluded that fluorodeoxyglucose, C-11 methionine, and F-18-fluoroethyl-L-tyrosine PET scans have pooled sensitivities of 63%, 94%, and 88%, respectively, and pooled specificities of 89%, 55%, and 57%, respectively, for determining the histopathologic grade of glioma.(97) **(EG 1)** The type of PET scan radiotracer used may make a difference in terms of diagnostic performance.(98)(99)(100) **(EG 2)** A study of 56 patients with resected glioblastoma compared follow-up imaging with C-11 methionine PET and F-18 fluorodeoxyglucose PET and found that C-11 methionine PET was associated with higher diagnostic accuracy for distinguishing between tumor recurrence and radiation necrosis compared with F-18 fluorodeoxyglucose PET.(101) **(EG 2)** An expert consensus guideline recommends brain MRI as the gold standard for diagnosis and staging; however, F-18 fluorodeoxyglucose PET scan is useful for distinguishing tumor recurrence from radiation injury and/or "pseudoprogression" due to concomitant treatment with temozolomide.(93) **(EG 2)** A prospective longitudinal study of 79 patients with newly diagnosed glioblastoma found that biological tumor volume identified by F-18-fluoroethyl-L-tyrosine PET imaging prior to radiochemotherapy was predictive of progression-free and overall survival.(102) **(EG 2)** A prospective study of 10 patients with recurrent high-grade glioma found that F-18-fluoroethyl-L-tyrosine PET scan was able to predict antiangiogenic treatment failure and thus contributed additional information for clinical management above that afforded by MRI alone.(103) **(EG 2)** A review article notes that C-11 methionine PET imaging is useful for the evaluation of primary gliomas and can play a role in differentiating between tumor recurrence and radiation necrosis.(104) **(EG 2)**

For symptoms suggestive of Parkinson disease or other neurodegenerative disorders, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Assessment of presynaptic dopaminergic activity using PET in conjunction with measurement of F-18 fluorodopa uptake and decarboxylation or assessment of dopamine transporter activity with one of several tracers has been used to differentiate between Parkinson disease and disorders that are not associated with presynaptic dopaminergic terminal deficiency (eg, essential tremor, vascular parkinsonism).(112)(113)(114) **(EG 2)** Expert consensus guidelines state that normal functional neuroimaging of the presynaptic dopaminergic system, which can be assessed by PET, is one of the exclusion criteria for diagnosis of Parkinson disease in individuals with parkinsonism (ie, bradykinesia in combination with rest tremor, rigidity, or both).(115)(116) **(EG 2)** A specialty society guideline states that presynaptic dopaminergic imaging (ie, with PET or SPECT) is indicated to differentiate between neurodegenerative parkinsonian syndromes and essential tremor and between idiopathic Parkinson disease and other forms of parkinsonism, but notes that it is not useful for distinguishing between idiopathic Parkinson disease and progressive supranuclear palsy, corticobasal degeneration, and subtypes of multiple system atrophy.(110) **(EG 2)** A systematic review of 4 studies concluded that there is limited evidence to support using F-18 fluorodeoxyglucose PET scan in the evaluation of patients with idiopathic Parkinson disease, in discriminating idiopathic Parkinson disease from progressive supranuclear palsy, and in identifying the neuropathologic cause of corticobasal syndrome. However, an expert panel still recommended the clinical use of F-18 fluorodeoxyglucose PET scan in the workup of patients with parkinsonism and dementia.(117) **(EG 1)** A specialty society guideline states that F-18 fluorodeoxyglucose PET may be appropriate for the evaluation of patients with parkinsonian syndromes (eg, Parkinson disease, multiple system atrophy), particularly for differentiating between idiopathic Parkinson disease and progressive supranuclear palsy, but notes that the evidence for F-18 fluorodeoxyglucose PET is limited.(118) **(EG 2)** Specialty society and expert consensus guidelines include F-18 fluorodeoxyglucose PET as part of the diagnostic evaluation of progressive supranuclear palsy and corticobasal syndrome.(111)(119) (120) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For brain metastasis from primary cancers, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** F-18 fluorodeoxyglucose PET scan is less sensitive than MRI of the brain and is therefore generally not recommended for this indication.(20) **(EG 2)** A systematic review and meta-analysis of 20 studies (including 728 patients with 872 brain metastases) evaluating the diagnostic performance of PET scan and MRI for differentiating tumor recurrence from radiation necrosis after stereotactic radiosurgery for brain metastases found pooled sensitivity and specificity of 84% and 86%, respectively, for PET scan and 84% and 88%, respectively, for MRI. However, the authors noted that most of the included studies were single-arm, moderate-quality studies and that there was heterogeneity among the studies in PET tracers used; further studies were recommended.(21) **(EG 1)** For detection of brain metastasis from non-small cell lung cancer, F-18 fluorodeoxyglucose PET scan performs poorly when compared with

either MRI or CT scan, with sensitivity of only approximately 40%.(22) **(EG 2)** An evidence-based clinical practice guideline on the treatment of small cell lung cancer notes that F-18 fluorodeoxyglucose PET scan is inferior to MRI or CT scan for the detection of brain metastases.(23) **(EG 2)** An expert consensus guideline concluded that F-18 fluorodeoxyglucose PET scan is inadequate for detecting brain metastasis from small cell lung cancer, as compared with MRI.(24) **(EG 2)**

For chronic traumatic encephalopathy (CTE), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review of 7 studies (403 patients) evaluating neuroimaging modalities for the diagnosis of CTE concluded that tau-binding PET is the most promising modality for distinguishing CTE from other neurologic diseases. However, due to the low specificity of diagnosing CTE premortem and low numbers of patients in the included studies, the authors recommended larger studies identifying suspected CTE based on published clinical criteria.(25) **(EG 1)** Review articles note that imaging studies such as PET are being used for research purposes, and further study is required to correlate imaging with clinical findings.(26)(27)(28) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(121); 42 CFR 422.101(122)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(123); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(124); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(125)

Medicare Coverage Determinations: Medicare Coverage Database(126)

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Footnotes

[A] Focal seizures show either clinical or EEG evidence of seizure onset from a localized area in the cerebral cortex.(84)(85) [A in Context Link 1]

Codes

CPT®: 78608, 78609, 78814

HCPCS: A9552, A9586, A9598, A9601, A9602, A9609, Q9982, Q9983, S8085

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